

✓ Oasis Infobyte Internship

Data Analytics Internship

Task 3: Sales Prediction using Linear Regression

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Domain: Data Analytics

Organization: Oasis Infobyte

Objective

The objective of this project is to predict product sales based on advertising expenditure using Linear Regression. By analyzing the relationship between advertising channels and sales, businesses can make better marketing decisions and optimize their advertising budget.

✓ 1. Importing Required Libraries

```
import pandas as pd
import numpy as np
import matplotlib.pyplot as plt
import seaborn as sns

from sklearn.model_selection import train_test_split
from sklearn.linear_model import LinearRegression
from sklearn.metrics import mean_absolute_error, mean_squared_error, r2_score

%matplotlib inline
```

```
!wget -O advertising.csv https://raw.githubusercontent.com/selva86/datasets/master/Advertising.csv
```

```
--2026-07-25 10:37:41-- https://raw.githubusercontent.com/selva86/datasets/master/Advertising.csv
Resolving raw.githubusercontent.com (raw.githubusercontent.com)... 185.199.109.133, 185.199.111.133
Connecting to raw.githubusercontent.com (raw.githubusercontent.com)|185.199.109.133|:443... connected
HTTP request sent, awaiting response... 200 OK
Length: 4756 (4.6K) [text/plain]
Saving to: 'advertising.csv'
```

```
advertising.csv 100%[=====] 4.64K --.-KB/s in 0s
```

```
2026-07-25 10:37:41 (33.0 MB/s) - 'advertising.csv' saved [4756/4756]
```

✓ 2. Load the Dataset

```
data = pd.read_csv("advertising.csv")
data
```

	Unnamed: 0	TV	radio	newspaper	sales	
0	1	230.1	37.8	69.2	22.1	
1	2	44.5	39.3	45.1	10.4	
2	3	17.2	45.9	69.3	9.3	
3	4	151.5	41.3	58.5	18.5	
4	5	180.8	10.8	58.4	12.9	
...	...	...	...	...	...	
195	196	38.2	3.7	13.8	7.6	
196	197	94.2	4.9	8.1	9.7	
197	198	177.0	9.3	6.4	12.8	
198	199	283.6	42.0	66.2	25.5	
199	200	232.1	8.6	8.7	13.4	

200 rows × 5 columns

Next steps:

[Generate code with data](#)[New interactive sheet](#)

3. Display First Five Rows

data.head()

	Unnamed: 0	TV	radio	newspaper	sales	
0	1	230.1	37.8	69.2	22.1	
1	2	44.5	39.3	45.1	10.4	
2	3	17.2	45.9	69.3	9.3	
3	4	151.5	41.3	58.5	18.5	
4	5	180.8	10.8	58.4	12.9	

Next steps:

[Generate code with data](#)[New interactive sheet](#)

4. Dataset Information

data.info()

```
<class 'pandas.core.frame.DataFrame'>
RangeIndex: 200 entries, 0 to 199
Data columns (total 5 columns):
#   Column      Non-Null Count  Dtype
---  -
0   Unnamed: 0  200 non-null   int64
1   TV          200 non-null   float64
2   radio       200 non-null   float64
3   newspaper   200 non-null   float64
4   sales       200 non-null   float64
dtypes: float64(4), int64(1)
memory usage: 7.9 KB
```

5. Dataset Shape

```
print("Dataset Shape:", data.shape)
```

Dataset Shape: (200, 5)

6. Missing Values

```
data.isnull().sum()
```

```

      0
Unnamed: 0  0
      TV      0
      radio  0
newspaper  0
      sales  0

dtype: int64

```

7. Duplicate Values

```
print("Duplicate Rows:", data.duplicated().sum())
```

Duplicate Rows: 0

8. Statistical Summary

```
data.describe()
```

	Unnamed: 0	TV	radio	newspaper	sales
count	200.000000	200.000000	200.000000	200.000000	200.000000
mean	100.500000	147.042500	23.264000	30.554000	14.022500
std	57.879185	85.854236	14.846809	21.778621	5.217457
min	1.000000	0.700000	0.000000	0.300000	1.600000
25%	50.750000	74.375000	9.975000	12.750000	10.375000
50%	100.500000	149.750000	22.900000	25.750000	12.900000
75%	150.250000	218.825000	36.525000	45.100000	17.400000
max	200.000000	296.400000	49.600000	114.000000	27.000000

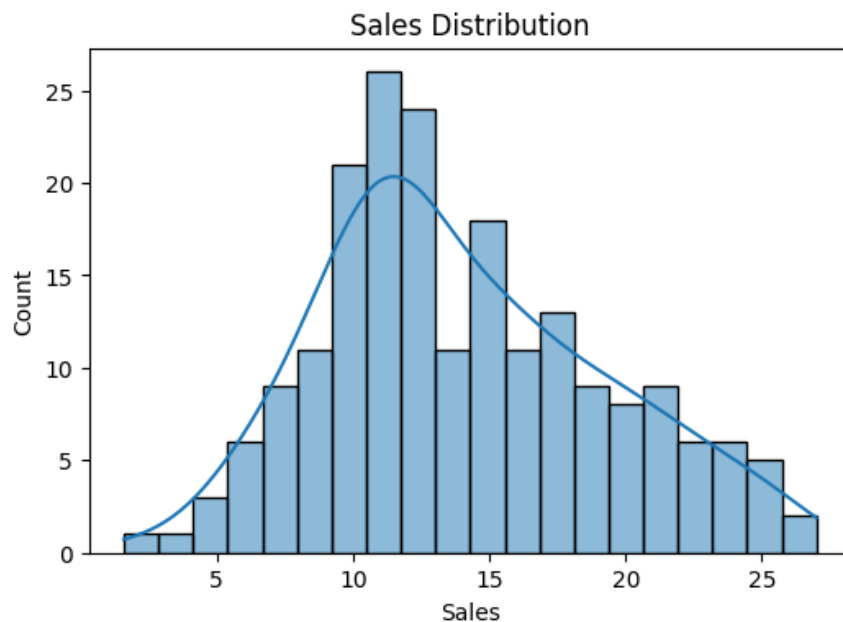
9. Exploratory Data Analysis (EDA)

9.1 Sales Distribution

```

plt.figure(figsize=(6,4))
sns.histplot(data['sales'], bins=20, kde=True)
plt.title('Sales Distribution')
plt.xlabel('Sales')
plt.ylabel('Count')
plt.show()

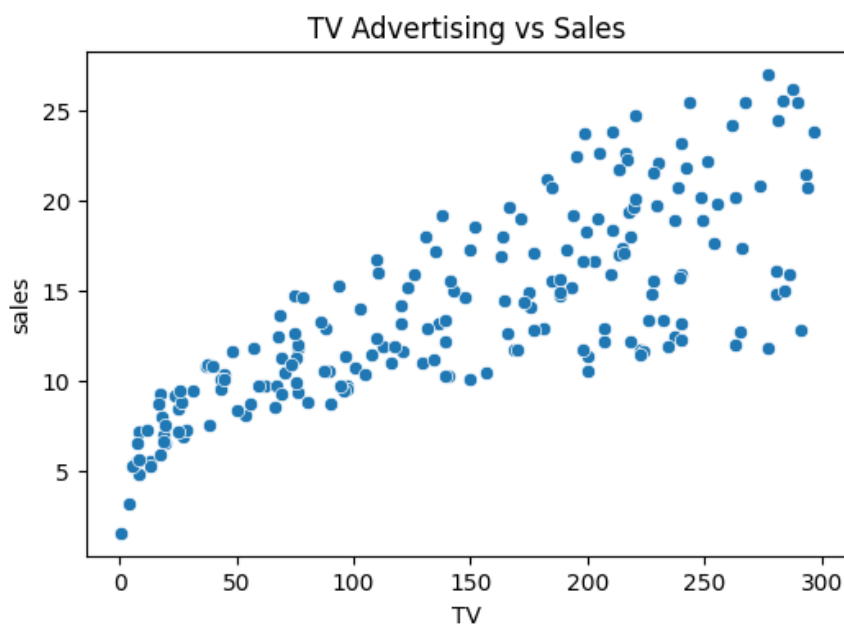
```

**Observation:**

- Sales values are distributed across different ranges.
- Most sales values are concentrated around the middle range.
- The distribution appears approximately normal with slight variation.

✓ 9.2 TV Advertising vs Sales

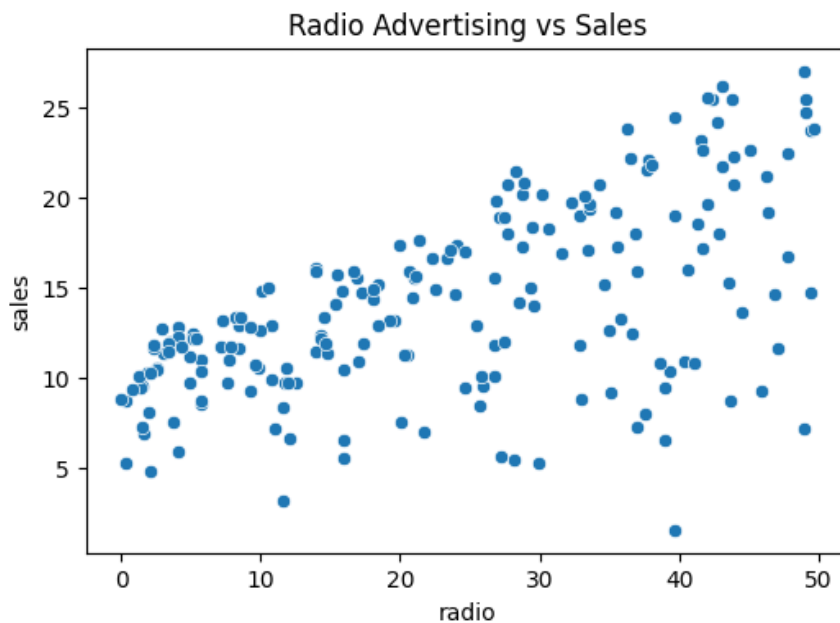
```
plt.figure(figsize=(6,4))
sns.scatterplot(x='TV', y='sales', data=data)
plt.title('TV Advertising vs Sales')
plt.show()
```

**Observation:**

- Sales increase as TV advertising budget increases.
- There is a strong positive relationship between TV advertising and sales.

✓ 9.3 Radio Advertising vs Sales

```
plt.figure(figsize=(6,4))
sns.scatterplot(x='radio', y='sales', data=data)
plt.title('Radio Advertising vs Sales')
plt.show()
```

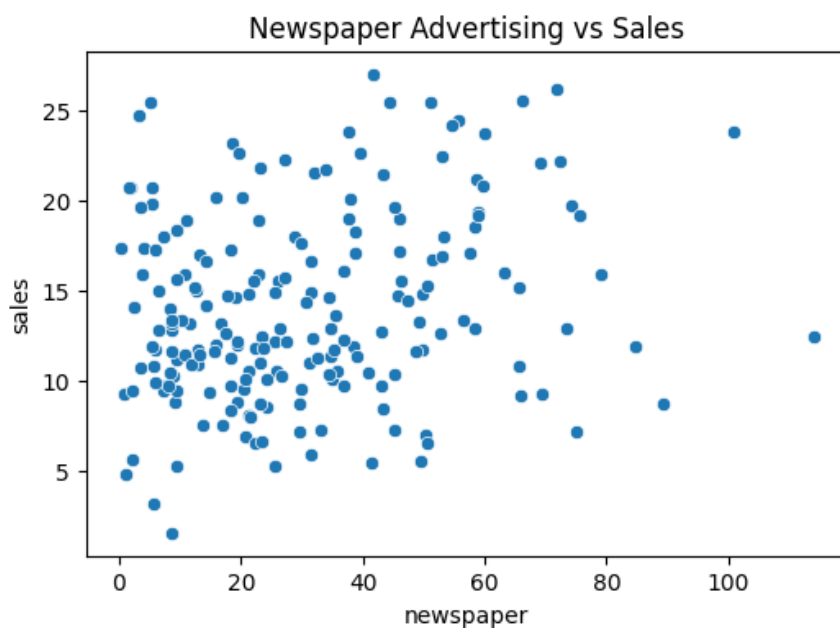


Observation:

- Radio advertising also shows a positive relationship with sales.
- The relationship is moderate compared to TV advertising.

✓ 9.4 Newspaper Advertising vs Sales

```
plt.figure(figsize=(6,4))
sns.scatterplot(x='newspaper', y='sales', data=data)
plt.title('Newspaper Advertising vs Sales')
plt.show()
```

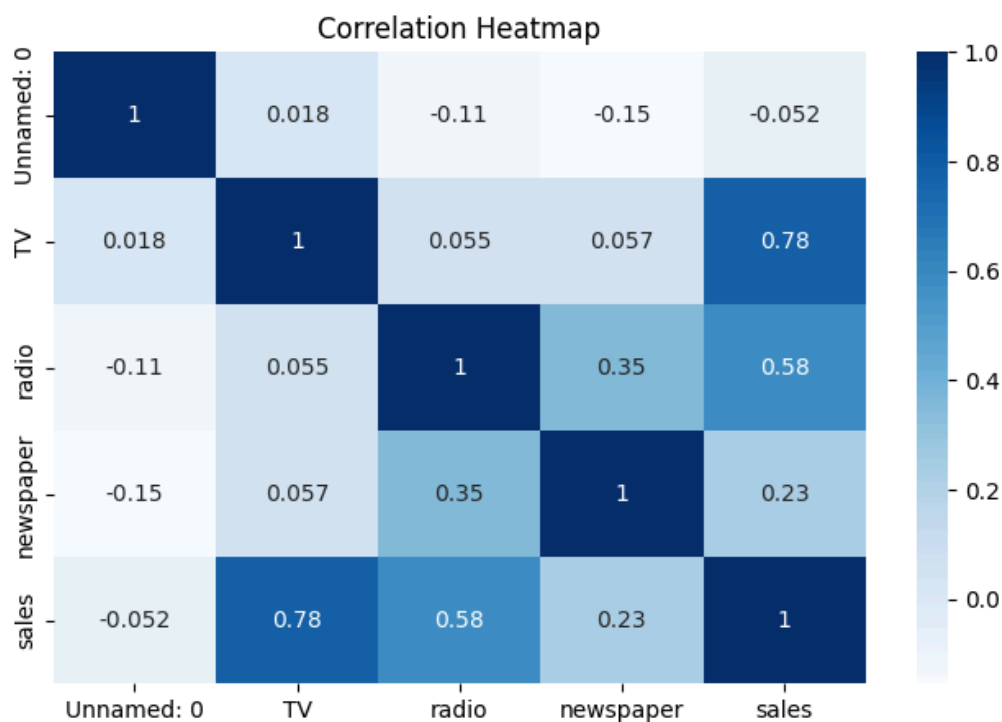


Observation:

- Newspaper advertising has a weak relationship with sales.
- Increasing newspaper advertising alone does not significantly improve sales.

✓ 9.5 Correlation Heatmap

```
plt.figure(figsize=(8,5))
sns.heatmap(data.corr(), annot=True, cmap='Blues')
plt.title("Correlation Heatmap")
plt.show()
```

**Observation:**

- TV has the strongest positive correlation with sales.
- Radio has a moderate positive correlation.
- Newspaper has the weakest correlation with sales.

✓ 10. Feature Selection

```
X = data[['TV', 'radio', 'newspaper']]
y = data['sales']
X.head()
```

	TV	radio	newspaper
0	230.1	37.8	69.2
1	44.5	39.3	45.1
2	17.2	45.9	69.3
3	151.5	41.3	58.5
4	180.8	10.8	58.4

Next steps: [Generate code with X](#) [New interactive sheet](#)

11. Train-Test Split

```
X_train, X_test, y_train, y_test = train_test_split(
    X, y, test_size=0.2, random_state=42
)
```

```
print("Training data shape:", X_train.shape)
print("Testing data shape:", X_test.shape)
```

```
Training data shape: (160, 3)
Testing data shape: (40, 3)
```

12. Model Building

```
model = LinearRegression()

model.fit(X_train, y_train)

print("Model trained successfully!")
```

```
Model trained successfully!
```

13. Model Prediction

```
y_pred = model.predict(X_test)

prediction = pd.DataFrame({
    'Actual Sales': y_test,
    'Predicted Sales': y_pred
})

prediction.head(10)
```

	Actual Sales	Predicted Sales	
95	16.9	16.408024	
15	22.4	20.889882	
30	21.4	21.553843	
158	7.3	10.608503	
128	24.7	22.112373	
115	12.6	13.105592	
69	22.3	21.057192	
170	8.4	7.461010	
174	11.5	13.606346	
45	14.9	15.155070	

Next steps: [Generate code with prediction](#) [New interactive sheet](#)

✓ 14. Model Evaluation

```
mae = mean_absolute_error(y_test, y_pred)
mse = mean_squared_error(y_test, y_pred)
rmse = np.sqrt(mse)
r2 = r2_score(y_test, y_pred)

print("Mean Absolute Error (MAE):", round(mae,2))
print("Mean Squared Error (MSE):", round(mse,2))
print("Root Mean Squared Error (RMSE):", round(rmse,2))
print("R² Score:", round(r2,2))
```

```
Mean Absolute Error (MAE): 1.46
Mean Squared Error (MSE): 3.17
Root Mean Squared Error (RMSE): 1.78
R² Score: 0.9
```

Observation:

- The Linear Regression model predicts sales with good accuracy.
- A low MAE and RMSE indicate that the prediction errors are small.
- The R² score shows how well the model explains the variation in sales.
- Overall, the model performs well for predicting sales based on advertising expenditure.

✓ 15. Actual vs Predicted Sales

```
plt.figure(figsize=(8,5))
plt.scatter(y_test, y_pred)
plt.xlabel("Actual Sales")
plt.ylabel("Predicted Sales")
plt.title("Actual vs Predicted Sales")
plt.show()
```

Observation:

- Most predicted values are close to the actual sales values.
- This indicates that the Linear Regression model fits the data well.

16. Conclusion

- Successfully built a Sales Prediction model using Linear Regression.
- Performed data preprocessing and exploratory data analysis.
- TV advertising showed the strongest influence on sales.
- Radio advertising had a moderate positive impact.
- Newspaper advertising had the least impact on sales.
- The model achieved good prediction accuracy with a high R² score.
- This model can help businesses optimize advertising budgets and improve sales forecasting.

